EXPERIMENT No2: Parallel Circuits

Student Guide

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Objective:
In this experiment, you will learn about parallel circuits, how they work, and how they differ from series circuits. You will construct a simple parallel circuit and observe how adding or removing components affects the circuit's behavior.

Materials Needed:

- 1 Breadboard (optional for neatness)
- 1 Battery (Power Source)
- 2 Light Bulbs (or LEDs)
- 3 Wires
- 1 Switch (optional for controlling the circuit)
- Resistors (if needed to limit current)

What is a Parallel Circuit?

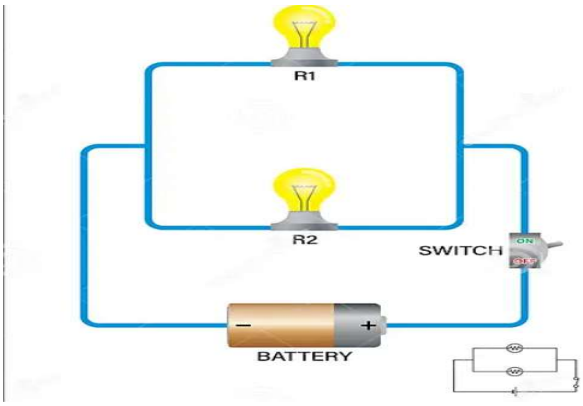
A parallel circuit is a type of electrical circuit in which all components are connected across the same two points, forming multiple paths for current to flow. In a parallel circuit:

- The voltage across each component is the same.
- The total current is the sum of the currents through each path.

Example: If you have two light bulbs in a parallel circuit, each light bulb will have the full voltage of the battery, and turning one off will not affect the other.

Steps to Construct a Parallel Circuit:

1) **<u>Schematic Diagram</u>**



2. Set up the Power Source:
   - Place the battery on the breadboard or connect it directly using wires.
3. Connect the First Light Bulb:
   a. Attach one terminal of the first light bulb to the positive terminal of the battery.
   b. Connect the other terminal of the light bulb to the negative terminal of the battery. The bulb should light up.
4. Add the Second Light Bulb in Parallel:
   a. Now, connect one terminal of the second light bulb to the same positive terminal of the battery.
   b. Connect the other terminal of the second light bulb to the same negative terminal of the battery. The second bulb should light up, and both bulbs will have the same brightness.

5. Test the Circuit:
    a. Observe the brightness of both bulbs. You will notice that both bulbs maintain their brightness even though they are sharing the same power source.
6. Optional: Add a Switch:
    a. Insert a switch in the circuit between the battery and the light bulbs so you can control when the circuit is on or off.

Observations:

- What happens to the brightness of the light bulbs when they are connected in parallel?
- Does turning off one light bulb affect the other?
- How does this compare to a series circuit where all components share the same path?

Theory Behind Parallel Circuits:

- In a parallel circuit, each component is connected directly to the power source, so the voltage across each component is the same.
- If one component fails or is removed, the other components in parallel continue to operate because each has its own independent path to the power source.
- The current through each component is determined by the resistance of the individual component, and the total current is the sum of the currents through each path.

Key Points:

- Voltage: The same across all components.
- Current: Splits between branches; total current is the sum of individual currents.
- Resistance: The total resistance in a parallel circuit is less than the resistance of the lowest-resistance branch.

Practical Applications:

Parallel circuits are commonly used in home wiring because they allow multiple devices to operate independently on the same power supply. For example, if one light goes out, the others will remain on.

Conclusion:

Through this experiment, you have learned how parallel circuits work, how they maintain the same voltage across all components, and how current is divided among the branches. Understanding parallel circuits is crucial in many real-life applications, such as in electrical devices and home appliances.

Challenge:

- Try adding a third light bulb to the circuit. What happens to the brightness of each bulb?
- What happens to the total current drawn from the battery as you add more bulbs?

This activity demonstrates the principles of parallel circuits and their applications in everyday electrical systems.